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Intro To Statistics

⇒ Statistics: It's the science of collecting, organizing and analyzing data.

⇒ Data: Facts or pieces of information

- ⇒ Appⁿ:
- ① data exploration & summarize
 - ② Model building & validation
 - ③ statistical analysis
 - ④ Hypothesis testing
 - ⑤ optimization & efficiency
 - ⑥ Reporting

Types of Statistics:

① Descriptive Statistics: It involves methods for summarizing & organizing data to make it understandable. This type of statistics help to describe the basic features of the data in a study.

⇒ Measure of central tendency

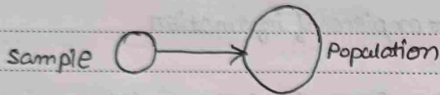
⇒ " " dispersion

⇒ data distribution (histogram, box plot etc.)

⇒ Summary statistics

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② Inferential Statistics: It involves methods for making predictions or inferences about a population based on sample of data. It allows for hypothesis testing, estimation & drawing conclusions.



⇒ Hypothesis testing
⇒ P-value

⇒ Confidence Interval

⇒ Statistical analysis test (Z-test, ~~ANOVA~~ Anova, chi-test)

Population & Sample Data:

⇒ Population: It's the entire set of individuals or objects of interest in a particular study.

⇒ Sample data: It's the subset of popⁿ that is used to represent the entire group.

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⇒ Characteristics of popⁿ:

① Complete set

② Parameter → a numeric value summarizing the entire popⁿ (e.g., popⁿ mean, variance etc)

⇒ e.g. of popⁿ:

① Popⁿ in a school study

→ all students in a school

⇒ charo. of sample data:

① subset of popⁿ

② statistic → a numerical value summarizing the sample data.

③ Random sampling: samples should be randomly selected to avoid bias.

Types of Sampling:

① Probability Sampling

② Non- " "

⇒ Types of Prob. Sampling:

① Simple Random Sampling: Every member of the popⁿ has an equal chance of being selected.

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e.g., selecting people randomly

(b) Systematic sampling: Select every n^{th} member of the popⁿ after a random starting point.

(c) Stratified sampling: divide the popⁿ into strata (groups) based on specific characteristic & then randomly sampling from each strata.
e.g.,

divide employees by dept & then randomly select a proportionate no. from each department to form a survey sample.

(d) Cluster sampling: divide the popⁿ into clusters & then randomly selecting clusters, then sampling all the members from the selected clusters.

(e) Multi-stage sampling: combining several sampling methods, usually involves selecting clusters then randomly sampling within those clusters.

⇒ Types of Non-Prob. Sampling:

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पुष्प स्टील इन्फ्रास्ट्रक्चर्स, पुष्प मकान बनारस

(a) convenience sampling: easiest to reach is selected

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(b) Judgemental / purposive sampling: selecting individuals based on the researcher's judgement.

(c) snowball sampling: existing study subjects recruit future subjects from among their acquaintances.

(d) Quota sampling: - based on quotas.

Types of Data:

⇒ Types of data:

Quantitative
(numerical)

Qualitative
(categorical)

→ Discrete
(whole No.)

→ Continuous
(any value)

→ Nominal
(No Ranking)

→ Ordinal
(have defined Ranks)

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Scales of Measurement of Data:

⇒ It describes the nature of info. within the values assigned to variables.

⇒ 4 Primary scale of measurement:

① Nominal Scale → it classifies data into distinct categories that don't have an intrinsic order.
characteristic:

② data is categorised based on label, name, or quality

③ categories are mutually exclusive.

④ No logical order among categories (i.e., Work) (i.e., Work)

② Ordinal scale → it classifies data into categories that can be ranked or ordered.

characteristics:

③ data is categorised & ranked in a specific order.

④ The interval b/w ranks are not necessarily equal.

③ Interval scale: → It not only categorises and orders but also specifies the exact diff. b/w intervals. It lacks a true zero point.

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characteristics:

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सुपर स्टील अस्पताल, सुपर माला कला

④ data is ordered with consistent interval b/w values

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⑤ allows for meaningful comparison of differences.

⑥ No true zero point.

④ Ratio scale →

⑤ order matters

⑥ contains a zero starting point

⑦ differences are measurable

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बालमुकुन्द सुपर स्टील

सुपर स्टील अस्पताल, सुपर माला कला